
MAD: Manifold Attracted Diffusion

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Abstract

Score-based diffusion models are a highly effective method for generating samples from a distribution of images. We consider scenarios where the training data comes from a noisy version of the target distribution, and present an efficiently implementable modification of the inference procedure to generate noiseless samples. Our approach is motivated by the manifold hypothesis, according to which meaningful data is concentrated around some low-dimensional manifold of a high-dimensional ambient space. The central idea is that noise manifests as low magnitude variation in off-manifold directions in contrast to the relevant variation of the desired distribution which is mostly confined to on-manifold directions. We introduce the notion of an extended score and show that, in a simplified setting, it can be used to reduce small variations to zero, while leaving large variations mostly unchanged. We describe how its approximation can be computed efficiently from an approximation to the standard score and demonstrate its efficacy on toy problems, synthetic data, and real data.

1. Introduction

Score-based diffusion models are a state-of-the-art approach for image synthesis, often outperforming alternatives like generative adversarial networks (GANs) and variational autoencoders (VAEs) in sample quality and diversity (Ho et al., 2020; Song & Ermon, 2020). These models learn a neural network by adding noise to data samples during training, according to some forward process. The network can then be used to reverse this process during inference. Essentially, they denoise an image with independent Gaussian pixel values into samples from the data distribution.

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However, real-world datasets are often corrupted by noise arising from measurement errors, compression artifacts, or data collection processes (Gupta & Gupta, 2019; Brummer & De Vleeschouwer, 2019). In this case, standard diffusion models trained directly on noisy data will learn to reproduce this corruption in their generated samples. We aim to address this by developing a method that can generate samples that approximately come from the clean distribution, despite being trained on noisy data.

A well-established paradigm for understanding data with a high-dimensional representation is the *manifold hypothesis*: meaningful data distributions are concentrated near a low-dimensional manifold embedded in the ambient space (Bengio et al., 2013; Fefferman et al., 2016). For instance, natural images, despite being represented as a d -dimensional array, where d can be in the millions, exhibit intrinsic dimensions that are far lower (Pope et al., 2021). This viewpoint will be the basis of our approach as we interpret noise as low-magnitude variations in directions orthogonal to the manifold, whereas meaningful variations of the underlying data correspond to movements along the manifold itself. The central idea is to exploit this geometric structure during the inference process to suppress the former while preserving the latter.

The manifold hypothesis has previously been considered in the context of diffusion models to study their convergence behavior, e.g. in (Bortoli, 2022), (Chen et al., 2023), (Tang & Yang, 2024), and (Potapchik et al., 2024). It has also been argued in (Stanczuk et al., 2024) that diffusion models can be leveraged to estimate the intrinsic dimension of data manifolds. Some existing works tackle the problem of noisy training (Daras et al., 2023; Lu et al., 2025), but require adapting the training process. More precisely, the approach introduced in (Daras et al., 2023) modifies the training loss to take into account the (unknown) degradation process, while the method in (Lu et al., 2025) is based on a specific training process where a small percentage of the data is clean. Outside the generative setting, there are also traditional manifold denoising techniques predating diffusion models (Hein & Maier, 2006; Gong et al., 2010; Wang & Carreira-Perpinán, 2010; Fefferman et al., 2018; Faigenbaum-Golovin & Levin, 2023). However, none of these approaches are efficient in generating new samples from the clean distribution.

This work introduces *Manifold Attracted Diffusion (MAD)*, an efficiently implementable modification to the inference procedure of score-based models. We define the concept of an “extended score”, which coincides with the standard score when that exists, but is also well-defined for Dirac delta distributions. In fact, it treats them essentially as Gaussians with a certain non-zero variance. This property can be leveraged in the inference procedure to reduce small off-manifold variations to almost zero while leaving larger on-manifold variations mostly unchanged, resulting in a soft thresholding effect. Thereby our method implicitly “attracts” the generated samples towards a low-dimensional structure, effectively filtering out noise. Importantly, a suitable approximation to the extended score can be computed easily from an approximation to the standard score, which enables the use of established training methods as well as pretrained networks under some conditions that are, e.g., satisfied by the framework of (Karras et al., 2022). Our key contributions include:

- The formal definition and analysis of the extended score;
- An inference algorithm that reduces noise in generated samples without needing a special training procedure, making it compatible with established frameworks and pretrained models;
- Empirical validation on toy problems and real-world image data, such as FFHQ, AFHQ, ImageNet, and EMPIAR-11618 (cryo electron microscopy data).

We also distinguish our work from other diffusion-based approaches. Methods for posterior sampling, such as DPS (Chung et al., 2023; 2022), are designed to solve inverse problems (Daras et al., 2024), such as denoising a single given image. Our goal is different: we learn from a dataset of noisy images to produce new, clean samples from the underlying distribution.

This paper is structured as follows. In Section 2, we review the necessary background on score-based diffusion models. In Section 3, we formally introduce the extended score and analyze its properties. Section 4 details our proposed inference algorithm and, in Section 5, we present numerical experiments that validate our approach. The related code is available at <https://github.com/delbraechter/MAD>.

2. Background

We will be working with the probability flow ODE formulation of diffusion models, largely following the framework of (Karras et al., 2022). For more background and the connection to other diffusion model formulations we refer to (Karras et al., 2022) and the references therein.

Given a data distribution p_0 on $\mathbb{R}^d$ we consider the stochastic process $(X_\sigma)_{\sigma \in [0, \sigma_{\max}]}$ with $X_0 \sim p_0$ and

$$X_\sigma = X_0 + \mathcal{N}(0, \sigma^2 \mathbb{I}), \quad \sigma > 0.$$

The corresponding densities are given by

$$p_\sigma = p_0 * g_{\sigma^2}, \quad (1)$$

where $g_{\sigma^2}(x) = (2\pi\sigma^2)^{-d/2} \exp(-\frac{\|x\|^2}{2\sigma^2})$ is the density of $\mathcal{N}(0, \sigma^2 \mathbb{I})$. The central idea behind score-based diffusion models is to generate a sample $x_0 \sim X_0$ from a sample $x_{\sigma_{\max}} \sim X_{\sigma_{\max}}$, where $x_{\sigma_{\max}}$ is, in practice, approximated by simply sampling from $\mathcal{N}(0, \sigma_{\max}^2 \mathbb{I})$, as for a sufficiently large $\sigma_{\max}$ this should only introduce a negligible error. One way to achieve this is by evolving the ODE

$$dx_t = -\dot{\sigma}(t)\sigma(t)Sp_{\sigma(t)}(x_t)dt, \quad (2)$$

where $\sigma: [0, T] \rightarrow [0, \sigma_{\max}]$ is some noise schedule and the score operator is given by

$$S: P(\mathbb{R}^d) \rightarrow C(\mathbb{R}^d, \mathbb{R}^d), \quad p \mapsto \nabla_x \log p,$$

where $P(\mathbb{R}^d) := \{p \in C^1(\mathbb{R}^d, \mathbb{R}): \int_{\mathbb{R}^d} p(x) dx = 1, p(x) > 0 \forall x \in \mathbb{R}^d\}$ is the set of densities that are positive everywhere in $\mathbb{R}^d$. Note that, for $\sigma > 0$, $p_0 * g_{\sigma^2}$ is a density even if p_0 is not. The fact that the score may not be well-defined for p_0 , e.g. because the data distribution is supported on some lower-dimensional subset in $\mathbb{R}^d$, is avoided in practice by generating a sample x_δ with δ close to 0. It can be shown (Song et al., 2021; Karras et al., 2022) that evolving a sample $x_{t_1} \sim X_{t_1}$ from t_1 to t_2 according to (2) yields a sample $x_{t_2} \sim X_{t_2}$.

Of course, this is only useful provided that we have access to the score, which depends on the data distribution, and from which we usually have only a finite number of samples. Remarkably, it turns out that a useful approximation of the score can be learned by training a neural network on these samples. A common practice, motivated by numerical stability, is to learn the so-called denoiser function D , which is simply a shifted and scaled version of the score, i.e. $D(x, \sigma) = \sigma^2 Sp_\sigma(x) + x$. This denoiser function is then approximated by a neural network D_θ using a loss based on

$$\mathbb{E}_{y \sim X_0} \mathbb{E}_{\eta \sim \mathcal{N}(0, \sigma^2 \mathbb{I})} \|D_\theta(y + \eta) - y\|_2^2$$

which is minimized by the denoiser function. However, as the expectation over the data distribution must be replaced by the empirical expectation based on the available data samples, this loss has, in general, many global minima. Nonetheless, employing neural networks with an adapted U-Net architecture (Ronneberger et al., 2015; Ho et al., 2020; Dhariwal & Nichol, 2021) appears to introduce sufficient bias towards a good approximation of the score.

This approximation capability is primarily founded on the empirical observation that using it to generate images by evolving (2) (or related differential equations) produces realistic samples. An analytical description of how D_θ approximates D is still an open problem. As the purpose of the work is to introduce a novel inference procedure, we will simply assume that we can obtain a suitable approximation to the score by some established training method.

3. Concept: the Extended Score

Let $M(\mathbb{R}^d)$ denote the set of probability measures on $\mathbb{R}^d$.

Definition 3.1 (Extended score). For $d \in \mathbb{R}^d$, $p \in M(\mathbb{R}^d)$, $\gamma \in (0, \infty)$, let

$$H_\gamma : M(\mathbb{R}^d) \rightarrow C(\mathbb{R}^d, \mathbb{R}^d),$$

$$p \mapsto (1 + \gamma)S(p * g_\gamma) + \gamma \frac{d}{d\gamma} S(p * g_\gamma)$$

and

$$H_0 : \widetilde{M}(\mathbb{R}^d) \rightarrow \{f : \mathbb{R}^d \rightarrow \mathbb{R}^d\}, \quad p \mapsto \lim_{\gamma \rightarrow 0} H_\gamma p,$$

where

$$\widetilde{M}(\mathbb{R}^d) := \{p \in M(\mathbb{R}^d) : \lim_{\gamma \rightarrow 0} (H_\gamma p)(x) \in \mathbb{R}^d \forall x \in \mathbb{R}^d\}.$$

We first note that H_0 coincides with the score for probability distributions with density in $P(\mathbb{R}^d)$, which we can view as a subset of $M(\mathbb{R}^d)$ by identifying a density function $p \in P(\mathbb{R}^d)$ with the measure given by $p(A) = \int_A p(x)dx$ for $A \subseteq \mathbb{R}^d$ (see Section B for a proof).

Lemma 3.2. Let $p \in P(\mathbb{R}^d)$. Then $H_0 p = S p$.

It is, however, also well-defined for, e.g., Dirac delta measures. Specifically, let δ denote the Dirac delta at 0, then

$$H_\gamma \delta(x) = (1 + \gamma)S(g_\gamma)(x) + \gamma \frac{d}{d\gamma} S(g_\gamma)(x)$$

$$= -\frac{(1+\gamma)x}{\gamma} + \gamma \frac{d}{d\gamma} \left(-\frac{x}{\gamma}\right) = -x.$$

In particular it holds that $H_0 \delta(x) = -x$, i.e. we obtain¹ a function which, similar to the score for Gaussians, yields at each point a vector pointing towards the mode of the probability distribution. This generalizes to mixtures of Dirac delta distributions, where $H_0 p$ will point towards the location of the nearest Dirac delta in the mixture (see Section B for a proof).

¹Note that in this simple case, $H_\gamma \delta$ is already the same as $H_0 \delta$, which is not the case in more complicated scenarios, e.g. for mixtures of Dirac deltas.

Lemma 3.3. Let $n \in \mathbb{N}$, $\mu_1, \dots, \mu_n \in \mathbb{R}^d$, $c_1, \dots, c_n \in \mathbb{R}_+$ such that $\sum_i c_i = 1$ and $p = \sum_{i \in [n]} c_i \delta_{\mu_i}$. Then

$$H_0 p(x) = - \sum_{i \in [n]} z_i(x)(x - \mu_i),$$

where $W_i = \{x \in \mathbb{R}^d : \|x - \mu_i\| \leq \|x - \mu_j\| \forall j \in [n]\}$ is a Voronoi region and

$$z_i(x) = \begin{cases} 0 & x \notin W_i, \\ 1 & x \in \text{int } W_i, \\ c_i \left(\sum_{j: x \in W_j} c_j \right)^{-1} & x \in \partial W_i. \end{cases}$$

The first two cases in the expression for $z_i(x)$ cover almost every (w.r.t the Lebesgue measure) $x \in \mathbb{R}^d$ and the third case is only needed if x is equally distant to multiple μ_i . By combining the expressions of the extended score for a Dirac delta and for a non-degenerate Gaussian distribution, it is possible to derive an explicit expression of the extended score for any (possibly degenerate) Gaussian distribution. This is due to how the extended score behaves with respect to products of measures, see Appendix A for the details.

We have the rather peculiar property that $H_0 \delta = S g_1$, i.e. the extended score of the Dirac delta matches the score of a variance 1 Gaussian². At this point, it should be noted that it is impossible to find a nice extension of the score operator that includes Dirac delta distributions, as

$$S g_\gamma(x) = -\frac{x}{\gamma},$$

which diverges for $x \neq 0$ as $\gamma \rightarrow 0$. As such any extension is necessarily discontinuous with respect to any topology in which $\lim_{\gamma \rightarrow 0} g_\gamma = \delta$.

However, the ability of the extended score to treat distributions with positive variance like they were Dirac delta distributions will, in fact, be the cornerstone of our proposed inference technique. Under the manifold hypothesis, the clean data distribution has significant variation only along a small number of directions. Given noise in the ambient pixel space, which has a much higher dimension than the image manifold, the variance in off-manifold directions due to noise should be much smaller (e.g. for isotropic Gaussian noise η the variance in each direction is of order $\sim d^{-\frac{1}{2}} \mathbb{E}[\|\eta\|]$). Thus, noise can be suppressed by using the extended score. This principle is formally justified in Appendix A, where we derive the extended score for a distribution supported on a low-dimensional subspace and show

²Changing the factor $(1 + \gamma)$ to $(1 + \frac{\gamma}{v})$ in the definition of H_γ would lead to the equivalency $H_0 \delta = S g_v$ instead of $H_0 \delta = S g_1$. We will exploit the fact that this equivalency is adjustable in the next section, albeit only implicitly as we also need to account for the fact that we cannot take the $\gamma \rightarrow 0$ limit in practice.

that it strongly attracts samples in the off-manifold directions while preserving the standard score in the on-manifold directions.

It is worth mentioning that the idea of decomposing the score function along the tangential and normal directions to the manifold was already exploited in (Chen et al., 2023) (in the case of linear subspaces) and in (Liu et al., 2026) (in the case of general manifolds). Several interesting theoretical aspects are discussed in these papers, which, however, do not deal with the denoising task discussed in this work.

4. Implementation

In view of the property of the extended score just discussed, we would like to design an inference procedure that is able to generate samples with less noise, if compared to the samples obtained via the usual score. To this end we first note that, due to (1), we have

$$\begin{aligned} S(p_{\sigma(t)} * g_\gamma) &= S(p_0 * g_{\sigma(t)^2} * g_\gamma) \\ &= S(p_0 * g_{\sigma(t)^2 + \gamma}) \\ &= S(p \sqrt{\sigma(t)^2 + \gamma}). \end{aligned} \quad (3)$$

So, given a network trained to approximate the score for any $\sigma \in [0, \sigma_{\max}]$, as is the case in the framework of (Karras et al., 2022), we also have an approximation to the extended score with a given small $\gamma > 0$, where the γ -derivative can be obtained, e.g., by a finite difference³ approximation. As the desirable properties of the extended score hold in the $\gamma \rightarrow 0$ limit, which we cannot compute directly, we instead need to choose a suitable dependence $\gamma(t)$ with $\lim_{t \rightarrow 0} \gamma(t) = 0$.

A basic way to conduct inference with the standard score is evolving (2) simply via Euler method, i.e. initializing with $x_0 \sim \mathcal{N}(0, \sigma(t_0)^2 \mathbb{I})$ and iterating

$$x_{i+1} = x_i - (t_{i+1} - t_i) \dot{\sigma}(t_i) \sigma(t_i) S_\theta(\sigma(t_i), x_i), \quad (4)$$

where $S_\theta(\sigma, x) \approx S_{p_\sigma}(x)$ is the learned approximation of the score. We will instead, based on Definition 3.1, consider the iteration

$$x_{i+1} = x_i - m(t_i)(t_{i+1} - t_i) \dot{\sigma}_\gamma(t_i) \sigma_\gamma(t_i) \alpha_i, \quad (5)$$

where

$$\alpha_i = ((1 + \gamma(t_i)) S_\theta(\sigma_\gamma(t_i), x_i) + \gamma(t_i) \frac{d}{d\gamma} S_\theta(\sigma_\gamma(t_i), x_i)),$$

$\sigma_\gamma(t) = \sqrt{a\sigma(t)^2 + b\gamma(t)}$, and $m(t_i)$ is a correction factor determined as explained below.

³Note that one could also compute the exact derivative of the score network using automatic differentiation (AD). However, since PyTorch is not optimized for forward mode AD this is significantly more expensive than a second evaluation of the network and did not yield any clear improvements in the numerical experiments.

Here, $a, b > 0$ are manually chosen parameters. Note that this reduces to (4) for $a = 1$, $b = 0$, and $m(t) = (1 + \gamma(t))^{-1}$, so essentially the choices of a , b , and m determine to what extent we would like the inference to push points onto a manifold at a given time step. More precisely, increasing a or reducing b corresponds to lessening the effect of the extended score and thereby limiting the denoising effect. Conversely, reducing a or increasing b yield a stronger denoising effect. As such, a and b can be seen as regularization parameters, and their choice is problem dependent. While in the idealized case of the $\gamma \rightarrow 0$ limit, analyzed in Section 3, Diracs and variance 1 Gaussians are treated in the same way, the choice of a and b adjusts this equivalence in practice, causing variations below this (soft) threshold to be compressed significantly. Note that we currently do not determine the explicit dependence of this threshold on the parameters. However, since we cannot expect to know the optimal threshold for a data distribution in practice, a parameter optimization seems unavoidable anyway. We further enforce $\sigma_\gamma(t) = t$ to match the scheduling of (Karras et al., 2022), as it allows us to take advantage of their optimized choice of time steps and leads to better comparability. Lastly, we introduce a parameter $p > 0$ to regulate the relative speed of convergence of σ and γ via $\gamma(t) = \sigma(t)^p$.

With these choices, (5) simplifies to

$$x_{i+1} = x_i - m(t_i)(t_{i+1} - t_i) t_i \left((1 + \gamma(t_i)) S_\theta(t_i, x_i) + \frac{b\gamma(t_i)}{2t_i} \frac{d}{d\sigma} S_\theta(t_i, x_i) \right), \quad (6)$$

where $\frac{d}{d\sigma} S_\theta$ denotes the derivative of $S_\theta: [0, \sigma_{\max}] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ w.r.t its first argument, and $\gamma(t_i)$ is a solution of

$$\sqrt{a\gamma(t_i)^{\frac{2}{p}} + b\gamma(t_i)} = t_i.$$

We observe that in the case of $S_\theta(\sigma, x) = -\frac{x-\mu}{\sigma^2}$, i.e. an initial distribution consisting of a Dirac delta at μ , the standard score inference rule (4) with $\sigma(t) = t$ evaluates to

$$\begin{aligned} x_{i+1} &= x_i - \frac{t_i - t_{i+1}}{t_i} (x_i - \mu) \\ &= (1 - \frac{t_i - t_{i+1}}{t_i}) x_i + \frac{t_i - t_{i+1}}{t_i} \mu. \end{aligned} \quad (7)$$

This means that, if $\frac{t_i - t_{i+1}}{t_i} \geq \beta > 0$ for every i , we have

$$\|x_{i+k} - \mu\| \leq (1 - \beta)^k \|x_i - \mu\|,$$

i.e. convergence to μ at a geometric rate. We would like our inference rule to mimic this behavior for simple Dirac deltas, which is accomplished by choosing the correction factor as

$$m(t_i) = \left(1 + \gamma(t_i) - \frac{b\gamma(t_i)}{t_i^2} \right)^{-1}, \quad (8)$$

see Section B in the appendix for a detailed derivation. Note that, as long as $\gamma(t_i) \in o(t_i^{-2})$ we have $\lim_{t \rightarrow 0} m(t) = 1$. Putting it all together, we arrive at the following algorithm.

Algorithm 1 Inference with extended score

```

1: function INFERENCE( $S_\theta, a, b, p, \Delta, (t_i)_{i \in \{0, \dots, N\}}$ )
2: sample  $x_0 \sim \mathcal{N}(0, t_0^2 \mathbb{I})$  // Generate Gaussian sample
3: for  $i \in \{0, \dots, N-1\}$  do
4:    $\gamma_i = \text{solve}(a\gamma_i^{2/p} + b\gamma_i - t_i^2 = 0)$  // Determine  $\gamma_i$  using a root finding algorithm
5:    $s_i = S_\theta(t_i, x_i)$  // Evaluate the score network
6:    $\tilde{s}_i = S_\theta((1 + \Delta)t_i, x_i)$ 
7:    $s'_i = \frac{\tilde{s}_i - s_i}{\Delta \cdot t_i}$  // Compute an approximation to the derivative
8:    $m_i = (1 + \gamma_i - \frac{b\gamma_i}{t_i^2})^{-1}$  // Compute correction factor
9:    $x_{i+1} = x_i - m_i(t_{i+1} - t_i)t_i((1 + \gamma_i)s_i + \frac{b\gamma_i}{2t_i}s'_i)$  // Update sample
10: end for
11: return  $x_N$ 
    
```

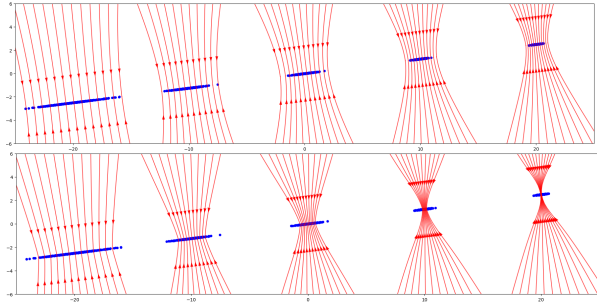


Figure 1. Samples (blue) of 1-dimensional Gaussian mixture supported on an affine subspace with variances 2, 0.5, 0.25, 0.1, and 0.05 respectively. Inference trajectories for standard score (top) and extended score (bottom) are indicated in red.

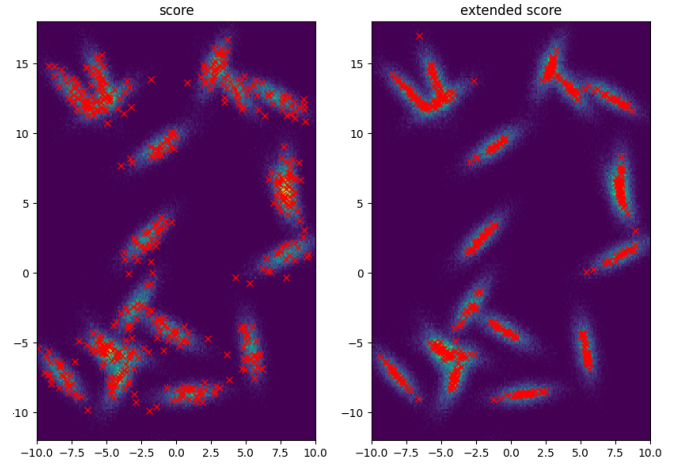
5. Numerical Experiments

In section 5.1 (toy examples in $\mathbb{R}^2$) and in section 5.2 (FFHQ, AFHQv2, and ImageNet) we present some illustrative numerical simulations to build intuition of the “manifold attraction” property of the extended score on clean datasets. Further, in sections 5.3 (synthetic dataset) and 5.4 (CIFAR-10) we provide controlled experiments showing qualitatively and quantitatively (FID scores) the denoising effect of the extended score. Finally, in section 5.5 we test MAD with real Cryo-EM data.

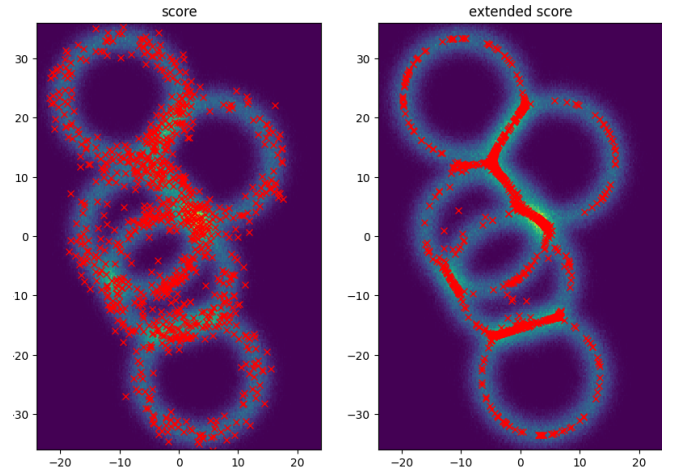
We emphasize that MAD addresses a specific blind generative denoising task where only noisy data is available, and the specific degradation model may be unknown. In this context, there are no established benchmarks, and a comparison with other (non-generative) manifold denoising approaches would not be meaningful, because they tackle a different problem.

5.1. Illustrative Examples in $\mathbb{R}^2$

We visualize the evolution according to the extended score in $\mathbb{R}^2$ for relatively simple distributions, where the (extended) score can be computed explicitly (see Appendix A). Figure 1



(a) Tilted Gaussians



(b) Radial Gaussians

Figure 2. Target distribution is displayed as a histogram colormap. Red crosses indicate samples generated via standard (left) or extended (right) score inference. Extended score parameters are $p = 1.3, a = 1, b = 1.1$ for (a) and $p = 2, a = 2, b = 30$ for (b).

displays the inference trajectories for a degenerate Gaussian mixture, which is supported on an 1-dimensional affine subspace, i.e. it behaves like a Dirac in the off-manifold direction. The means are chosen such that the different mixture components affect each other very little. We can see that for the Gaussians with low variance using the extended score causes all trajectories to end up at the respective means, i.e. it behaves as if we had Dirac deltas (in the on-manifold direction) at these locations. For the higher variance Gaussians the extended score trajectories still end up closer to the mean than they would for standard score, but this variance reduction effect decreases significantly as the variance of the Gaussians increases. All together, we essentially have a soft thresholding effect, where variances below a certain value are shrunk to 0, while large enough variances are left almost unchanged.

In Figure 2a we consider a mixture of Gaussians with covariance matrices $\Sigma_i = R_i \text{diag}(1.7, 0.2)$, where the R_i , $i \in \{1, \dots, 21\}$, are randomly chosen rotation matrices, i.e. around each mean of the mixture we have large variance in one direction and small variance in the direction orthogonal to the first one. Locally this can be viewed as a 1.5-variance Gaussian along some 1-dimensional affine linear subspace to which Gaussian noise with covariance matrix $0.2\mathbb{I}$ is added, resulting in a 1.7 variance in one direction and a 0.2 variance in the other. The denoising effect is clear: the extended score inference moves the points onto the affine linear subspace corresponding to the first principal direction of each Gaussian, while leaving the spread along the affine linear subspace almost unchanged. In Figure 2b we demonstrate that this effect is not limited to affine linear manifolds, by considering a mixture of radial Gaussians of the form $p(x) = C \sum_{i=1}^5 \exp(-\frac{(\|x - \mu_i\| - r)^2}{2v})$ with variance $v = 2.5$, radius $r = 10$, randomly chosen means μ_i , and C a normalizing constant.

5.2. Effects on Noiseless FFHQ, AFHQv2, and ImageNet

While the last section provides some basic intuition for the effects of our inference method, it is, of course, unclear how this translates to much more complicated distributions in much higher dimensions. We will explore this question using our inference method with the pretrained score networks from (Karras et al., 2022) for⁴ FFHQ ((Karras et al., 2019)), AFHQv2 ((Choi et al., 2020)), and ImageNet ((Deng et al., 2009)), where we also use the time schedule suggested in (Karras et al., 2022), and only vary the hyperparameters specific to our method, i.e. a , b , p , and Δ in Algorithm 1. As these datasets are arguably essentially noiseless, the effect of the extended score will be visible on the different features of the images. The following sections consider noisy data.

⁴Note that these networks have been trained on images from these datasets which have been downsampled to 64×64 resolution.

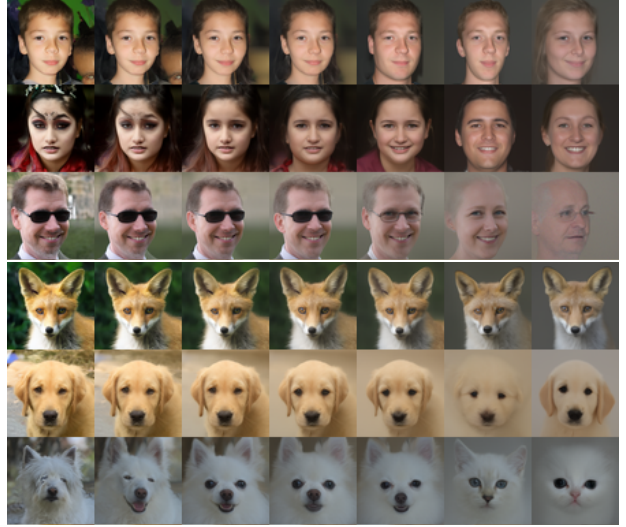


Figure 3. In both subfigures, all images in each row start from the same latent noise sample, and the leftmost column uses standard score whereas second to last columns use Algorithm 1 with $\Delta = 0.0001$, $p = 8$, $a = 2.5$, and, left to right, $b \in \{2, 5, 10, 20, 40, 80\}$. Further examples in Figure 13.

In Figure 3, we showcase the impact of our method on the generation of human or animal faces using the pretrained score networks for FFHQ and AFHQv2, respectively. Note that increasing the parameter b in Algorithm 1 results in a greater impact of the extended score, i.e. we expect stronger attraction to the manifold of primary variation. We observe that for all values of b , we generate samples with qualitatively the same facial features, but for larger values of b , we always get a plain single-colored background. In case of faces, it seems quite clear that the direction of primary relevance should correspond to essential facial features that are present in all the data, whereas the background variation, as well as features like glasses and head wear, are split across a much larger number of directions and are thresholded out first. Furthermore, as illustrated in the appendix (see Figures 12 and 13), the method demonstrates improved stability across hyperparameter variations on FFHQ compared to simpler synthetic datasets. This indicates that the higher complexity of natural images, reflected into the higher dimensionality of the underlying manifold, may inherently improve hyperparameter robustness. We also observe similar effects on ImageNet, e.g. in the examples in Figure 4, where the extended score inference seems to focus on generating one primary object, while progressively thresholding out everything else as we increase b . However, since ImageNet contains a large variety of objects, the primary directions differ between classes, and thus the effect of extended score inference with a given choice of hyperparameters is much more varied across different starting noise images (see Figure 14 in the Appendix).



Figure 4. Each row starts from the same latent noise sample, and the leftmost column uses standard score whereas second to last columns use Algorithm 1 with $\Delta = 0.001$, $p = 12$, and $a = 8$, $b \in \{5, 20, 50, 80, 250\}$ for the top row, $a = 4$, $b \in \{2, 10, 30, 60, 100\}$ for the bottom row.

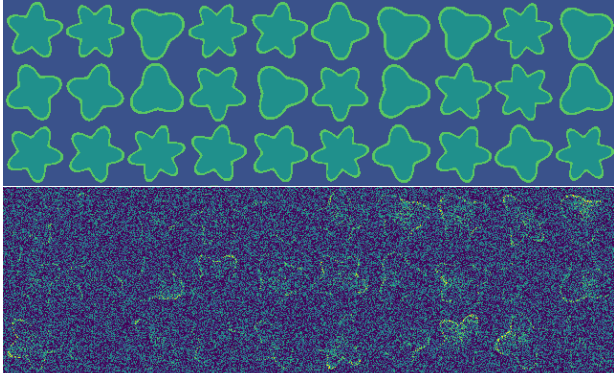


Figure 5. Grid of clean samples from the synthetic data set (top); grid of corresponding noisy samples (bottom). Each sample is an 8-bit grayscale 64×64 image, displayed via viridis colormap.

Although Algorithm 1 is deterministic when started with the same random seed, i.e. such that it starts from the same latent noise image, it relies on a score network trained through a highly stochastic process. Consequently, even small changes of the iterates x_i can build up and lead to the generation of a significantly different image. As can be seen in Figure 3, in general the algorithm is not particularly suited for orthogonal projection on the manifold of, in this case, faces, i.e. it does not simply generate the same face as standard inference would but without background. However, as showcased in Figure 12 in the Appendix, there are often certain ranges of parameters that lead to rather similar images and may be used for manual adjustment of a generated image.

5.3. Denoising Synthetic Data

We created a synthetic dataset of clean samples, added noise, and trained a diffusion model only on the noisy data. Our clean distribution consists of 8-bit grayscale 64×64 images depicting 4 different shapes, each appearing with equal probability and rotated by an angle chosen uniformly at random (see Figure 5, top). As such, this probability distribution is essentially supported on 4 disjoint 1-dimensional manifolds in pixel space. We then add two types of corruption to obtain noisy samples. First, we blot out large parts of

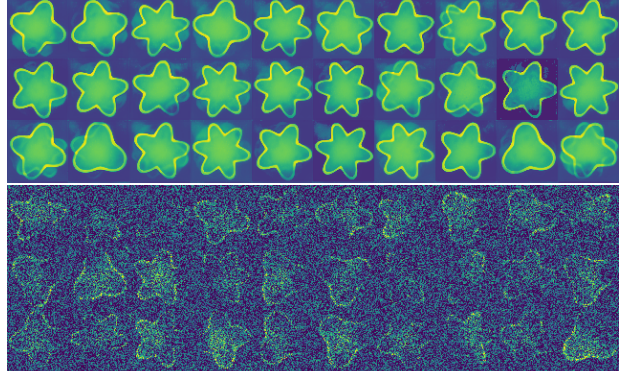


Figure 6. Grid of samples generated from consecutive random seeds by Algorithm 1 with parameters $\Delta = 0.02$, $p = 8$, $a = 0.002$, and $b = 15$ (top); grid of samples generated by inference with standard score (bottom). Each sample is an 8-bit grayscale 64×64 image, displayed via viridis colormap.

the shape, by uniformly randomly picking 50 locations on the boundary of the shape and subtracting Gaussian bump functions centered at those locations from the clean image. Second, we add i.i.d. Gaussian noise to each pixel, resulting in images where the original shapes can hardly be discerned from a single image (see Figure 5, bottom). We then trained a DDPM++ model⁵ on a set of 100 000 of noisy samples for a duration of 3 mimg, i.e. 30 repeats per image. When using the trained network for standard score inference we obtain a reproduction of the corrupted samples (see Figure 6, bottom). In contrast, running our extended score inference, we obtain samples showing the original shapes (see Figure 6, top). Figure 15 in the Appendix illustrates the dependence on the extended score parameters Δ , a , b , and p . In order to test whether our method works on corrupted training data beyond the case of Gaussian noise, we also trained a DDPM++ model, using the same training settings, on a set of 100 000 synthetic samples, where each pixel was set to 0 with probability 0.5 (see Figure 17 in Appendix C). As can be seen in Table 1, for both cases extended score inference leads to a significant improvement in FID compared to the standard score, which generates samples with essentially the same distance to the clean data as the noisy data.

5.4. Denoising CIFAR-10 and FFHQ

Next, we test whether the denoising capabilities of our method extend to real data. Specifically we train DDPM++ models on CIFAR-10 and FFHQ with additive Gaussian noise. While standard score inference produces image with roughly the amount of noise of the images it was trained on, extended score inference produces significantly cleaner images, both visually (Figure 7 and Figure 8) as well as

⁵Using the method from the github to (Karras et al., 2022) with default settings except for $\text{cond} = 0$ and $\text{augment} = 0$, which took roughly 14 hours on a single A100 GPU.

Noisy dataset	training data	standard generation	extended generation
synthetic data, Gaussian noise	320.56	319.42	189.51
synthetic data, pixel removal	233.86	234.40	163.14
CIFAR-10, Gaussian noise	136.87	147.47	77.15
FFHQ, Gaussian noise	133.60	134.42	74.52
FFHQ, Shot noise	257.28	257.42	101.12
FFHQ, pixel removal	372.21	387.87	104.00

Table 1. FIDs with respect to the corresponding clean dataset, computed from 10 000 images.

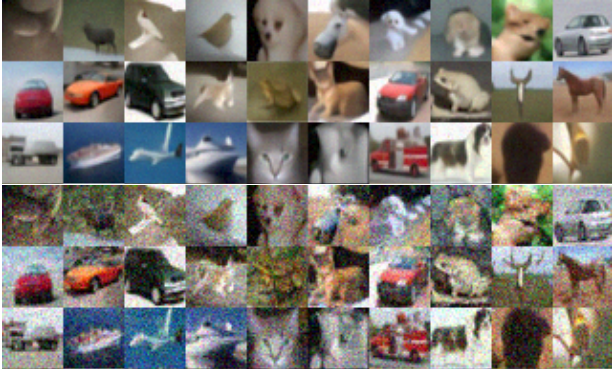


Figure 7. Standard score inference (bottom) and extended score inference (top) for CIFAR-10 with additive Gaussian noise.

quantitatively in the form of FID scores (Table 1). We note that the FID values for standard generation very closely match the values for the respective training sets, indicating that in training the diffusion model simply learns to reproduce the corruption and does not contribute to the denoising effect. The intermediate images generated during inference (see Figure 18 in Appendix C) strongly indicate that this effect cannot simply be achieved by truncated sampling.

We also train DDPM++ models on FFHQ corrupted by shot/Poisson noise (Figure 9) or pixel removal noise (Figure 10). While the images generated by extended score inference are of somewhat worse quality than in the Gaussian case, the FID values of the training sets are significantly larger, i.e. we are dealing with a higher degree of corruption. In fact, MAD achieves a larger FID improvement over standard generation than in the Gaussian case. More importantly, it demonstrates that MAD is effective for non-Gaussian corruptions, which is evidence that it does not simply modify the amount of Gaussian noise removed during inference.

5.5. Denoising Real Data

In this section we test our method on real data from single-particle Cryo-Electron Microscopy (see (Cheng et al., 2015) for an introduction), where many particles of the same type are suspended in liquid, frozen, and put under an electron microscope. This produces extremely noisy 2D-images which need to be refined before further steps like 3D recon-

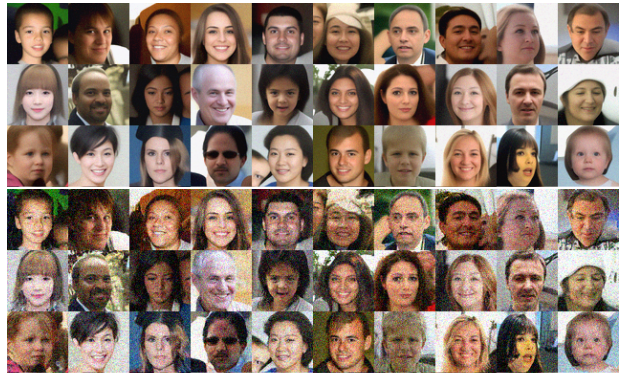


Figure 8. Standard score inference (bottom) and extended score inference (top) for FFHQ with additive Gaussian noise.

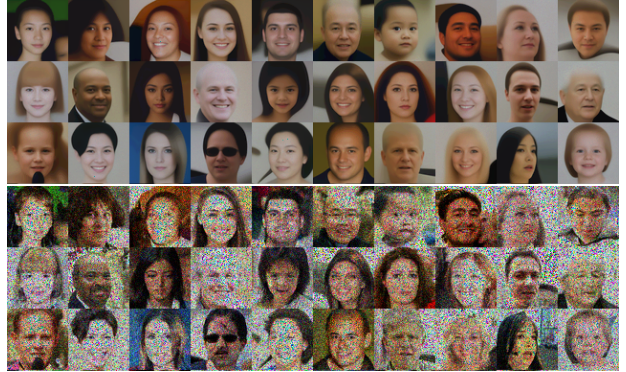


Figure 9. Standard score inference (bottom) and extended score inference (top) for FFHQ with Shot/Poisson noise.

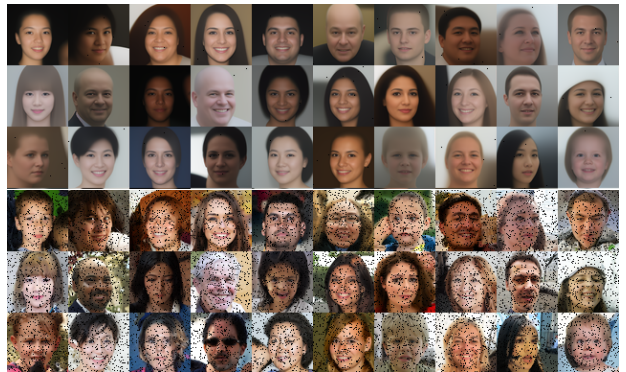


Figure 10. Standard score inference (bottom) and extended score inference (top) for FFHQ with pixel removal noise.

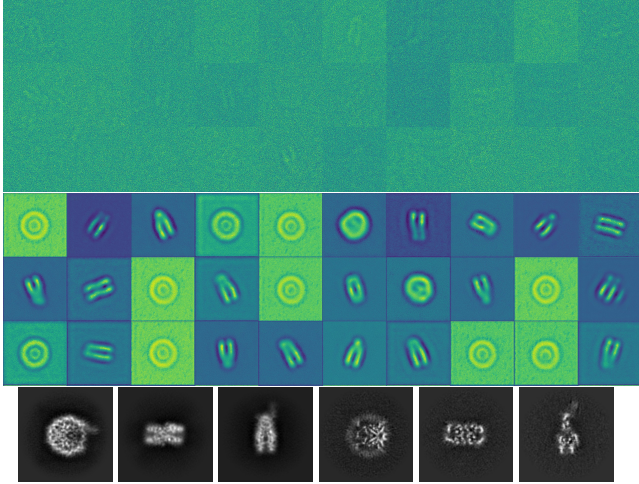


Figure 11. Images from the EMPIAR-11618 dataset (top). Samples generated by Algorithm 1 with $\Delta = 0.001$, $p = 8$, $a = 0.01$, $b = 2$ (middle). Shapes obtained by (Bacic et al., 2021) (bottom), taken from (wwPDB Consortium, 2023, EMD-17944). The colormap is applied with normalization per image to enhance contrast.

struction can be attempted. This presents an opportunity to investigate the performance of our method on practically relevant real world data with non-Gaussian noise. We use the EMPIAR-11618 (Bacic et al., 2021) dataset of 68 401 grayscale images with 256×256 resolution. They were extracted from raw data and undergone some preprocessing, but are still very noisy (see Figure 11, top). We trained a DDPM++ model⁶ on this data and used our method to generate samples (see Figure 11, middle) whose shapes correspond strongly to what has been obtained by (Bacic et al., 2021), see Figure 11, bottom. We emphasize that the network has only ever seen noisy data and has in no way been specifically adjusted based on a priori knowledge of these shapes. This can be seen by the fact that standard score inference generates noisy samples, very similar to those in the training set (see Figure 19 in the Appendix). The parameters used for Figure 11 of the extended score inference have been determined by hand with such knowledge, of course, but similar results are generated for a wide range of parameter choices (see Figure 16 in the Appendix). In particular, this demonstrates a significant capability of the extended score to guide generation towards samples from the underlying image manifold, also in the case of real data with extreme noise corruption caused by physical measurement modalities. This example serves as proof of concept for the validation of our approach: reaching state-of-the-art performance is outside the scope of this work, and would require incorporating more prior domain knowledge.

⁶In order to compensate for the higher resolution we reduced the number of feature channels in the DDPM++ architecture from 128 to 32, but otherwise used the same settings as for the synthetic data. Training took roughly 60 hours on two A100 GPUs.

6. Conclusion

We introduced Manifold Attracted Diffusion (MAD), a novel inference approach for score-based diffusion models to generate clean samples from a distribution despite training on noisy datasets. Based on the manifold hypothesis, our method utilizes the underlying geometry to suppress off-manifold variations while preserving on-manifold variations, which results in attracting samples toward a low-dimensional structure. It can leverage established training algorithms and pretrained models. The required computation cost is at most twice that of standard inference, due to requiring a second evaluation of the score network. However, we observed empirically that it is enough to use the extended score only in the last half of the inference steps, yielding an additional computational cost of 50%, but additional investigation on this aspect is required. Numerical experiments on both synthetic and real data demonstrate that MAD successfully suppresses noise.

Future directions include extending MAD to solve inverse problems, as in Cryo-EM for denoising individual images, or for other denoising or image restoration tasks. This would benefit from integration with conditional diffusion models (Chung et al., 2023; 2022). Developing automatic and adaptive parameter selection for $\gamma(t)$ would enhance the methods robustness and applicability. Further theoretical analysis of the extended score may lead to an improved incorporation into the inference procedure, in particular one could combine it with inference techniques using noise injection or higher order ODE discretization schemes. Future work should also involve a rigorous quantitative study comparing the performance and computational trade-offs of MAD against training-time methods, finetuning on limited clean data, and other relevant inference-time baselines such as truncated sampling (Daras et al., 2025). Furthermore, it would be valuable to investigate whether the MAD framework could be adapted to time-unconditioned generative models (Sun et al., 2025), or if our method fundamentally relies on a time-dependent score. It may also be interesting to consider whether the MAD framework can be adapted for Variance Preserving diffusion models. The key difficulty would be that one cannot use identity (3) in that setting. Finally, a promising future direction, motivated by the conceptual similarities to classifier-free guidance (Ho & Salimans, 2021), is to explore whether our extended score can serve as a more interpretable, geometrically-grounded method for improving generation quality, even for models trained on clean data.

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Conflict of Interest

The authors declare no conflict of interest.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

References

- Bacic, L., Gaullier, G., Sabantsev, A., Lehmann, L. C., Brackmann, K., Dimakou, D., Halic, M., Hewitt, G., Boulton, S. J., and Deindl, S. Structure and dynamics of the chromatin remodeler alc1 bound to a parylated nucleosome. *eLife*, 10:e71420, sep 2021. ISSN 2050-084X. doi: 10.7554/eLife.71420. URL <https://doi.org/10.7554/eLife.71420>.
- Bengio, Y., Courville, A., and Vincent, P. Representation learning: A review and new perspectives. *IEEE transactions on pattern analysis and machine intelligence*, 35(8): 1798–1828, 2013.
- Bortoli, V. D. Convergence of denoising diffusion models under the manifold hypothesis. *Transactions on Machine Learning Research*, 2022. ISSN 2835-8856. URL <https://openreview.net/forum?id=MhK5aXo3gB>.
- Brummer, B. and De Vleeschouwer, C. Natural image noise dataset. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, June 2019.
- Chen, M., Huang, K., Zhao, T., and Wang, M. Score approximation, estimation and distribution recovery of diffusion models on low-dimensional data. In Krause, A., Brunskill, E., Cho, K., Engelhardt, B., Sabato, S., and Scarlett, J. (eds.), *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 4672–4712. PMLR, 23–29 Jul 2023. URL <https://proceedings.mlr.press/v202/chen23o.html>.
- Cheng, Y., Grigorieff, N., Penczek, P. A., and Walz, T. A primer to single-particle cryo-electron microscopy. *Cell*, 161:438–449, 2015. doi: 10.1016/j.cell.2015.03.050.
- Choi, Y., Uh, Y., Yoo, J., and Ha, J.-W. Stargan v2: Diverse image synthesis for multiple domains. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2020.
- Chung, H., Sim, B., Ryu, D., and Ye, J. C. Improving diffusion models for inverse problems using manifold constraints. *Advances in Neural Information Processing Systems*, 35:25683–25696, 2022.
- Chung, H., Kim, J., Mccann, M. T., Klasky, M. L., and Ye, J. C. Diffusion posterior sampling for general noisy inverse problems. In *The Eleventh International Conference on Learning Representations*, 2023.
- Daras, G., Shah, K., Dagan, Y., Gollakota, A., Dimakis, A., and Klivans, A. Ambient diffusion: Learning clean distributions from corrupted data. *Advances in Neural Information Processing Systems*, 36:288–313, 2023.
- Daras, G., Chung, H., Lai, C.-H., Mitsufuji, Y., Ye, J. C., Milanfar, P., Dimakis, A. G., and Delbracio, M. A survey on diffusion models for inverse problems. *CoRR*, abs/2410.00083, 2024. URL <https://doi.org/10.48550/arXiv.2410.00083>.
- Daras, G., Cherapanamjeri, Y., and Daskalakis, C. C. How Much is a Noisy Image Worth? Data Scaling Laws for Ambient Diffusion. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=qZwtPEw2qN>.
- Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., and Fei-Fei, L. Imagenet: A large-scale hierarchical image database. In *2009 IEEE Conference on Computer Vision and Pattern Recognition*, pp. 248–255, 2009. doi: 10.1109/CVPR.2009.5206848.
- Dhariwal, P. and Nichol, A. Diffusion models beat gans on image synthesis. In Ranzato, M., Beygelzimer, A., Dauphin, Y., Liang, P., and Vaughan, J. W. (eds.), *Advances in Neural Information Processing Systems*,

- volume 34, pp. 8780–8794. Curran Associates, Inc., 2021. URL https://proceedings.neurips.cc/paper_files/paper/2021/file/49ad23d1ec9fa4bd8d77d02681df5cfa-Paper.pdf.
- Faigenbaum-Golovin, S. and Levin, D. Manifold reconstruction and denoising from scattered data in high dimension. *J. Comput. Appl. Math.*, 421:Paper No. 114818, 24, 2023. ISSN 0377-0427,1879-1778. doi: 10.1016/j.cam.2022.114818. URL <https://doi.org/10.1016/j.cam.2022.114818>.
- Fefferman, C., Mitter, S., and Narayanan, H. Testing the manifold hypothesis. *Journal of the American Mathematical Society*, 29(4):983–1049, 2016.
- Fefferman, C., Ivanov, S., Kurylev, Y., Lassas, M., and Narayanan, H. Fitting a putative manifold to noisy data. In *Conference On Learning Theory*, pp. 688–720. PMLR, 2018.
- Gong, D., Sha, F., and Medioni, G. Locally linear denoising on image manifolds. In *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics*, pp. 265–272. JMLR Workshop and Conference Proceedings, 2010.
- Grafakos, L. *Classical Fourier analysis*, volume 249 of *Graduate Texts in Mathematics*. Springer, New York, second edition, 2008. ISBN 978-0-387-09431-1.
- Gupta, S. and Gupta, A. Dealing with noise problem in machine learning data-sets: A systematic review. *Procedia Computer Science*, 161:466–474, 2019. ISSN 1877-0509. doi: <https://doi.org/10.1016/j.procs.2019.11.146>. URL <https://www.sciencedirect.com/science/article/pii/S1877050919318575>. The Fifth Information Systems International Conference, 23-24 July 2019, Surabaya, Indonesia.
- Hein, M. and Maier, M. Manifold denoising. In Schölkopf, B., Platt, J., and Hoffman, T. (eds.), *Advances in Neural Information Processing Systems*, volume 19. MIT Press, 2006. URL https://proceedings.neurips.cc/paper_files/paper/2006/file/a0b83c02d720415dada82e08bc09e9f3-Paper.pdf.
- Ho, J. and Salimans, T. Classifier-free diffusion guidance. In *NeurIPS 2021 Workshop on Deep Generative Models and Downstream Applications*, 2021. URL <https://openreview.net/forum?id=qw8AKxfYbI>.
- Ho, J., Jain, A., and Abbeel, P. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, volume 33, pp. 6840–6851, 2020.
- Karras, T., Laine, S., and Aila, T. A style-based generator architecture for generative adversarial networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2019.
- Karras, T., Aittala, M., Aila, T., and Laine, S. Elucidating the design space of diffusion-based generative models. In *Advances in Neural Information Processing Systems*, volume 35, pp. 26565–26577, 2022.
- Liu, Z., Zhang, W., and Li, T. Improving the euclidean diffusion generation of manifold data by mitigating score function singularity. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2026. URL <https://openreview.net/forum?id=T98sh6P0Vm>.
- Lu, H., Wu, Q., and Yu, Y. Stochastic forward-backward deconvolution: Training diffusion models with finite noisy datasets. In *Forty-second International Conference on Machine Learning*, 2025. URL <https://openreview.net/forum?id=WrwQv3mpQx>.
- Pope, P., Zhu, C., Abdelkader, A., Goldblum, M., and Goldstein, T. The intrinsic dimension of images and its impact on learning. In *International Conference on Learning Representations*, 2021. URL <https://openreview.net/forum?id=XJk19XzGq2J>.
- Potapchik, P., Azangulov, I., and Deligiannidis, G. Linear convergence of diffusion models under the manifold hypothesis. *arXiv preprint arXiv:2410.09046*, 2024.
- Ronneberger, O., Fischer, P., and Brox, T. U-net: Convolutional networks for biomedical image segmentation. In Navab, N., Hornegger, J., Wells, W. M., and Frangi, A. F. (eds.), *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*, pp. 234–241, Cham, 2015. Springer International Publishing. ISBN 978-3-319-24574-4.
- Song, Y. and Ermon, S. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations*, 2020.
- Song, Y., Sohl-Dickstein, J., Kingma, D. P., Kumar, A., Ermon, S., and Poole, B. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations*, 2021. URL <https://openreview.net/forum?id=PXTIG12RRHS>.
- Stanczuk, J. P., Batzolis, G., Deveney, T., and Schönlieb, C.-B. Diffusion models encode the intrinsic dimension of data manifolds. In *Forty-first International Conference on Machine Learning*, 2024.

Sun, Q., Jiang, Z., Zhao, H., and He, K. Is noise conditioning necessary for denoising generative models? In *Forty-second International Conference on Machine Learning*, 2025. URL <https://openreview.net/forum?id=pTSWi6RTtJ>.

Tang, R. and Yang, Y. Adaptivity of diffusion models to manifold structures. In Dasgupta, S., Mandt, S., and Li, Y. (eds.), *Proceedings of The 27th International Conference on Artificial Intelligence and Statistics*, volume 238 of *Proceedings of Machine Learning Research*, pp. 1648–1656. PMLR, 02–04 May 2024.

Wang, W. and Carreira-Perpinán, M. A. Manifold blurring mean shift algorithms for manifold denoising. In *2010 IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, pp. 1759–1766. IEEE, 2010.

wwPDB Consortium, T. Emdb—the electron microscopy data bank. *Nucleic Acids Research*, 52(D1):D456–D465, 11 2023. ISSN 0305-1048. doi: 10.1093/nar/gkad1019. URL <https://doi.org/10.1093/nar/gkad1019>.

A. The Extended Score for Products of Measures

In the following lemma, we derive the expression of the extended score of the product of two measures. In this section, the superscripts denote the dimension of the domain.

Lemma A.1. *Let $d_1 + d_2 = d$. Take $p_1 \in \widetilde{M}(\mathbb{R}^{d_1})$ and $p_2 \in \widetilde{M}(\mathbb{R}^{d_2})$. Then the product measure $p = p_1 \otimes p_2$ belongs to $\widetilde{M}(\mathbb{R}^d)$ and*

$$H_0^d p(x) = (H_0^{d_1} p_1(x_1), H_0^{d_2} p_2(x_2)),$$

where $x = (x_1, x_2) \in \mathbb{R}^{d_1+d_2}$.

Proof. We have that $g_\gamma^d(x) = g_\gamma^{d_1}(x_1)g_\gamma^{d_2}(x_2)$. Thus, by Fubini's theorem, we have

$$(p * g_\gamma^d)(x) = (p_1 * g_\gamma^{d_1})(x_1) \cdot (p_2 * g_\gamma^{d_2})(x_2).$$

Thus

$$\log(p * g_\gamma^d)(x) = \log(p_1 * g_\gamma^{d_1})(x_1) + \log(p_2 * g_\gamma^{d_2})(x_2).$$

Taking a gradient with respect to x we obtain

$$S^d(p * g_\gamma^d)(x) = (S^{d_1}(p_1 * g_\gamma^{d_1})(x_1), 0_{d_2}) + (0_{d_1}, S^{d_2}(p_2 * g_\gamma^{d_2})(x_2)).$$

Therefore, since the expression of $H_\gamma^d(p)$ is linear in $S^d(p * g_\gamma^d)$, we obtain

$$H_\gamma^d(p)(x) = (H_\gamma^{d_1}(p_1)(x_1), 0_{d_2}) + (0_{d_1}, H_\gamma^{d_2}(p_2)(x_2)) = (H_\gamma^{d_1}(p_1)(x_1), H_\gamma^{d_2}(p_2)(x_2)).$$

Taking the limit as $\gamma \rightarrow 0$, the result follows. $\square$

This result can be used to calculate the extended score of a degenerate distribution p , namely, a distribution supported on a lower-dimensional affine subspace of $\mathbb{R}^d$. Since the score, and thus the extended score, is equivariant with respect to rotations and translations, without loss of generality, we can assume that p is supported on $\{(x_1, 0_{d_2}) \in \mathbb{R}^d : x_1 \in \mathbb{R}^{d_1}\}$ with density $p_1 \in P(\mathbb{R}^{d_1})$, namely,

$$p = p_1 \otimes \delta_{d_2},$$

where $\delta_{d_2} \in \widetilde{M}(\mathbb{R}^{d_2})$ is the Dirac delta centered at 0 in $\mathbb{R}^{d_2}$. By Lemmata 3.2 and A.1, we obtain that the extended score of p is given by

$$H_0^d p(x) = (H_0^{d_1} p_1(x_1), H_0^{d_2} p_2(x_2)) = (S^{d_1} p_1(x_1), -x_2),$$

where we have also used that $H_0 \delta(x) = -x$.

In the particular case when p_1 is a non-degenerate Gaussian distribution on $\mathbb{R}^{d_1}$ with mean $\mu_1 \in \mathbb{R}^{d_1}$ and covariance $\Sigma_1 \in \mathbb{R}^{d_1 \times d_1}$, namely $p_1 = \mathcal{N}(\mu_1, \Sigma_1)$, we have

$$H_0^d p(x) = (-\Sigma_1^{-1}(x_1 - \mu_1), -x_2).$$

As expected, this coincides with the standard score of the (non-degenerate) Gaussian distribution $\mathcal{N}((\mu_1, 0_{d_2}), \Sigma)$ on $\mathbb{R}^d$, where Σ is the block matrix given by

$$\Sigma = \begin{bmatrix} \Sigma_1 & 0 \\ 0 & \mathbb{I}^{d_2} \end{bmatrix}.$$

B. Proofs and Derivations

Proof of Lemma 3.2. By standard properties of approximate identities (Grafakos, 2008, Example 1.2.17 and Theorem 1.2.19(2)), for $f \in C(\mathbb{R}^d)$ bounded we have that

$$\lim_{\gamma \rightarrow 0} (f * g_\gamma)(x) = f(x), \quad x \in \mathbb{R}^d,$$

Analogously, with $h_\gamma(x) := \frac{1}{\gamma} \|x\|^2 g_\gamma(x)$ by (Grafakos, 2008, Theorem 1.2.21(b)) we get that

$$\lim_{\gamma \rightarrow 0} (f * h_\gamma)(x) = Cf(x), \quad x \in \mathbb{R}^d,$$

where $C = \int_{\mathbb{R}^d} (2\pi)^{-\frac{d}{2}} \|z\|^2 e^{-\frac{\|z\|^2}{2}} dz < \infty$. In addition, we observe that

$$\frac{\partial}{\partial x_i} g_\gamma(x) = -\frac{x_i}{\gamma} g_\gamma(x)$$

and

$$\frac{d}{d\gamma} g_\gamma(x) = \left(-\frac{d}{2\gamma} + \frac{\|x\|^2}{2\gamma^2}\right) g_\gamma(x) = \frac{1}{2\gamma} (h_\gamma(x) - dg_\gamma(x)).$$

Consequently, as $p(x) > 0$ for all $x \in \mathbb{R}^d$, by assumption

$$\begin{aligned} \lim_{\gamma \rightarrow 0} \gamma \frac{d}{d\gamma} S(p * g_\gamma) &= \lim_{\gamma \rightarrow 0} \gamma \frac{d}{d\gamma} \frac{\nabla_x(p * g_\gamma)}{p * g_\gamma} = \lim_{\gamma \rightarrow 0} \gamma \frac{d}{d\gamma} \frac{\nabla_x p * g_\gamma}{p * g_\gamma} \\ &= \lim_{\gamma \rightarrow 0} \gamma \frac{(p * g_\gamma)(\nabla_x p * \frac{d}{d\gamma} g_\gamma) - (\nabla_x p * g_\gamma)(p * \frac{d}{d\gamma} g_\gamma)}{(p * g_\gamma)^2} \\ &= \lim_{\gamma \rightarrow 0} \frac{(p * g_\gamma)((\nabla_x p * h_\gamma) - d(\nabla_x p * g_\gamma)) - (\nabla_x p * g_\gamma)((p * h_\gamma) - d(p * g_\gamma))}{2(p * g_\gamma)^2} \\ &= \frac{p(C\nabla_x p - d\nabla_x p) - \nabla_x p(Cp - dp)}{2p^2} \\ &= 0. \end{aligned}$$

Moreover, it holds that

$$\lim_{\gamma \rightarrow 0} S(p * g_\gamma) = \lim_{\gamma \rightarrow 0} \frac{\nabla_x(p * g_\gamma)}{p * g_\gamma} = \frac{\lim_{\gamma \rightarrow 0} \nabla_x p * g_\gamma}{\lim_{\gamma \rightarrow 0} p * g_\gamma} = \frac{\nabla_x p}{p} = Sp,$$

which completes the proof. $\square$

Proof of Lemma 3.3. We write $h_i(x) := c_i e^{-\frac{\|x - \mu_i\|^2}{2\gamma}}$, i.e. $p * g_\gamma = \sum_{i \in [n]} (2\pi\gamma)^{-\frac{d}{2}} h_i$, and observe that

$$S(p * g_\gamma)(x) = - \sum_{i \in [n]} \frac{x - \mu_i}{\gamma} w_i(x),$$

where

$$w_i := \frac{h_i}{\sum_{j \in [n]} h_j}.$$

We observe that

$$\frac{d}{d\gamma} w_i(x) = \frac{(\sum_{j \in [n]} h_j(x)) \frac{\|x - \mu_i\|^2}{2\gamma^2} h_i(x) - h_i(x) \sum_{j \in [n]} \frac{\|x - \mu_j\|^2}{2\gamma^2} h_j(x)}{(\sum_{j \in [n]} h_j(x))^2}$$

and

$$H_\gamma p(x) = (1 + \gamma) S(p * g_\gamma)(x) + \gamma \frac{d}{d\gamma} S(p * g_\gamma)(x) = \gamma S(p * g_\gamma)(x) + \frac{d}{d\gamma} \gamma S(p * g_\gamma)(x).$$

We will first show that the second term vanishes for $\gamma \rightarrow 0$. To this end, we note

$$\begin{aligned} \frac{d}{d\gamma} \gamma S(p * g_\gamma)(x) &= - \sum_{i \in [n]} (x - \mu_i) \frac{(\sum_{j \in [n]} h_j(x)) \frac{\|x - \mu_i\|^2}{2\gamma^2} h_i(x) - h_i(x) \sum_{j \in [n]} \frac{\|x - \mu_j\|^2}{2\gamma^2} h_j(x)}{(\sum_{j \in [n]} h_j(x))^2} \\ &= - \frac{\sum_{i, j \in [n]} h_i(x) h_j(x) (x - \mu_i) (\|x - \mu_i\|^2 - \|x - \mu_j\|^2)}{2\gamma^2 (\sum_{j \in [n]} h_j(x))^2}. \end{aligned}$$

Let $x \in W_k$ and $i, j \in [n]$ such that $x \notin W_i \vee x \notin W_j$, then

$$\lim_{\gamma \rightarrow 0} \frac{h_i(x)h_j(x)}{\gamma^2(\sum_{j \in [n]} h_j(x))^2} \leq \lim_{\gamma \rightarrow 0} \frac{h_i(x)h_j(x)}{\gamma^2 h_k(x)^2} = \lim_{\gamma \rightarrow 0} \frac{c_i c_j}{\gamma^2 c_k^2} e^{\frac{2\|x - \mu_k\|^2 - \|x - \mu_i\|^2 - \|x - \mu_j\|^2}{2\gamma}} = 0$$

as $2\|x - \mu_k\|^2 - \|x - \mu_i\|^2 - \|x - \mu_j\|^2 < 0$ by definition of W_i . Since $\|x - \mu_i\|^2 - \|x - \mu_j\|^2 = 0$ if $x \in W_i \cap W_j$, and $h_i(x) \geq 0$ for every $x \in \mathbb{R}^d$, $i \in [n]$, we have

$$\lim_{\gamma \rightarrow 0} \frac{d}{d\gamma} \gamma S(p * g_\gamma)(x) = 0.$$

We proceed by noting that

$$\lim_{\gamma \rightarrow 0} \frac{h_j(x)}{h_i(x)} = \lim_{\gamma \rightarrow 0} \frac{c_j}{c_i} e^{-\frac{\|x - \mu_j\|^2 - \|x - \mu_i\|^2}{2\gamma}} = \begin{cases} 0, & \|x - \mu_i\| < \|x - \mu_j\| \\ \frac{c_j}{c_i}, & \|x - \mu_i\| = \|x - \mu_j\| \\ \infty, & \|x - \mu_i\| > \|x - \mu_j\| \end{cases}$$

and, consequently, using the conventions that $\frac{1}{0} = \infty$ and $\frac{1}{\infty} = 0$,

$$\begin{aligned} \lim_{\gamma \rightarrow 0} w_i(x) &= \lim_{\gamma \rightarrow 0} \frac{h_i(x)}{\sum_{j \in [n]} h_j(x)} \\ &= \lim_{\gamma \rightarrow 0} \left(1 + \sum_{j \in [n], j \neq i} \frac{h_j(x)}{h_i(x)} \right)^{-1} \\ &= \begin{cases} 0, & x \notin W_i \\ 1, & (x \in W_i) \wedge (x \notin W_j \forall j \in [n] \setminus \{i\}) \\ (\sum_{j \in J} \frac{c_j}{c_i})^{-1}, & i \in J \subseteq [n] : x \in \bigcap_{j \in J} W_j(x) \end{cases} \end{aligned}$$

Thus

$$H_0 p(x) = \lim_{\gamma \rightarrow 0} H_\gamma p(x) = \lim_{\gamma \rightarrow 0} \gamma S(p * g_\gamma)(x) = - \sum_{i \in [n]} (x - \mu_i) z_i(x).$$

This concludes the proof. $\square$

Derivation of equation (8). For $S_\theta(\sigma, x) = -\frac{x - \mu}{\sigma^2}$, we have

$$\frac{d}{d\sigma} S_\theta(\sigma, x) = \frac{2(x - \mu)}{\sigma^3}$$

and thus (6) evaluates to

$$\begin{aligned} x_{i+1} &= x_i - m(t_i)(t_{i+1} - t_i)t_i \left((1 + \gamma(t_i)) \left(-\frac{x_i - \mu}{t_i^2} \right) + \frac{b\gamma(t_i)}{2t_i} \frac{2(x_i - \mu)}{t_i^3} \right) \\ &= x_i - m(t_i) \frac{t_i - t_{i+1}}{t_i} (x_i - \mu) \left(1 + \gamma(t_i) - \frac{b\gamma(t_i)}{t_i^2} \right). \end{aligned}$$

We would like to force this to match the standard score inference step in (7), for this special case of $S_\theta(\sigma, x)$, which is achieved by choosing

$$m(t_i) = (1 + \gamma(t_i) - \frac{b\gamma(t_i)}{t_i^2})^{-1}.$$

$\square$



Figure 12. Generated with Algorithm 1 from the same latent noise sample with $\Delta = 0.0001$, $p = 8$, and, from left to right, $b \in \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ as well as, from top to bottom, $a \in \{1, 3, 5, 7\}$.

C. Additional Numerical Examples

Additional examples related to the datasets FFHQ and AFHQv2 are shown in Figure 13. Additional examples related to ImageNet are shown in Figure 14.

Figure 15 and Figure 16 show that the samples generated by extended score inference have a strong dependence on the chosen hyperparameters, but a significant emergence of structure can be observed for many different choices. Despite being a simpler problem, the generation seems to be less stable w.r.t. hyperparameter choices for the synthetic data. One reason for this might be that we have a 1-dimensional manifold of images that is harder to find than the one underlying the EMPIAR-11618 data, which is, of course, not explicitly known but can be assumed to be higher-dimensional. It may also be due to a shorter training duration (3ming compared to 10ming), which would be consistent with the generation being much more stable w.r.t. the hyperparameters for FFHQ, AFHQv2, and ImageNet as the networks for those problems have been trained significantly longer.

In Figure 17, we show the results with the synthetic dataset with 50% pixel removal. We see that images generated by standard score inference replicate the corruption present in the training set, while the images generated by extended score inference exhibit all key features of the clean data, while eliminating the corruption due to pixel removal.

In Figure 18, we compare the inference paths for standard and extended score generations: applying truncated sampling to the standard score generation would not be enough to obtain the denoising effect.

Additional images from the EMPIAR-11618 dataset, as well as images generated by standard score inference, are shown in Figure 19.

D. LLM use

LLMs were used to polish the writing for parts of the text, to suggest related work, and as a coding aid. All those suggestions have only been implemented after thorough manual review. No LLMs were involved in any way in the mathematical derivations.

⁷In particular, if $i = j$.

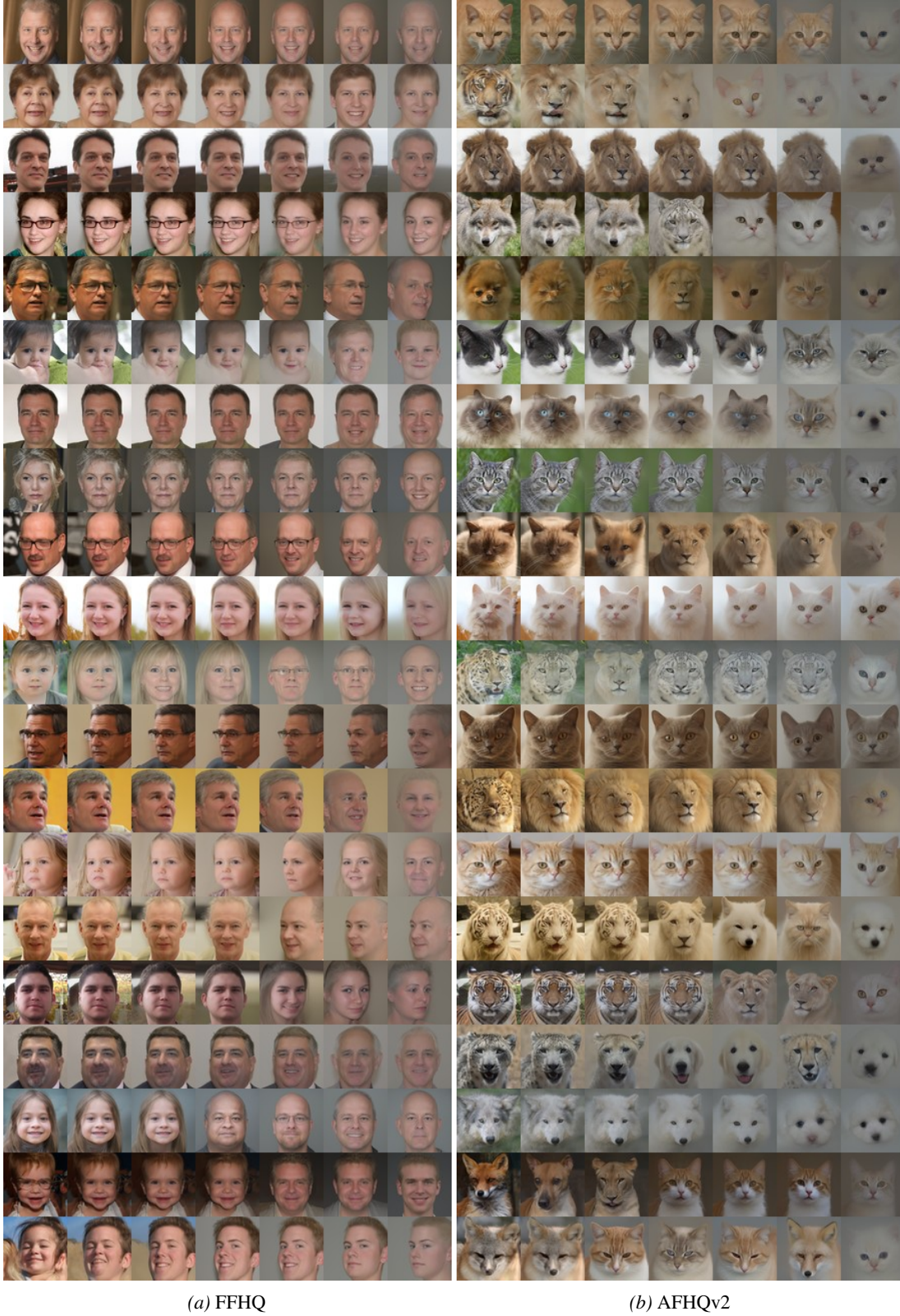


Figure 13. In both subfigures, all images in each row starts from the same latent noise sample and the leftmost column uses standard score whereas second to last columns use Algorithm 1 with $\Delta = 0.0001$, $p = 8$, $a = 2.5$, and, from left to right, $b \in \{2, 5, 10, 20, 40, 80\}$. The rows are generated from consecutive random seeds.

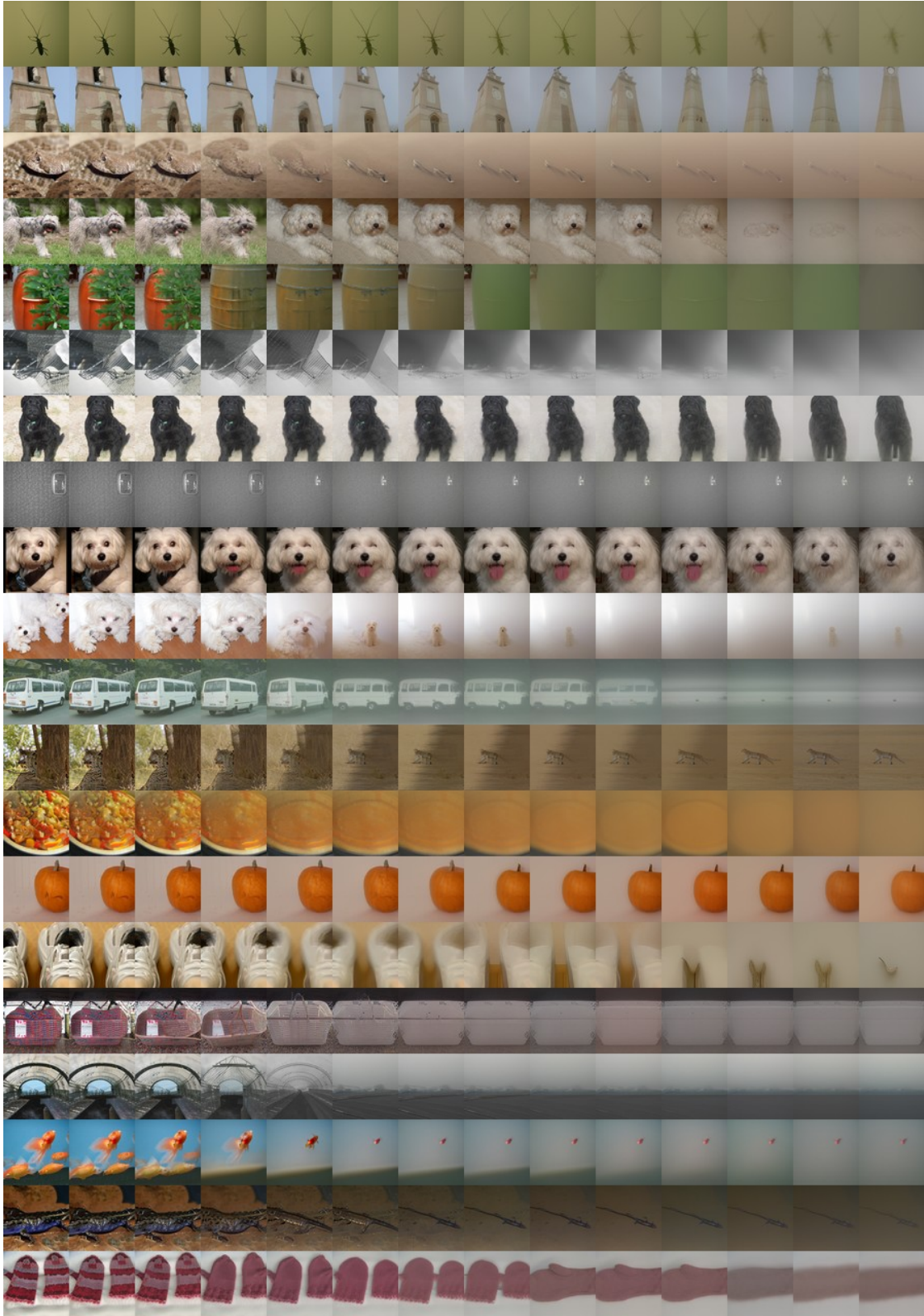


Figure 14. All images in each row starts from the same latent noise sample and the leftmost column uses standard score whereas second to last columns use Algorithm 1 with $\Delta = 0.001$, $p = 12$, $a = 4$, and, from left to right, $b \in \{1, 2, 5, 10, 15, 20, 25, 30, 35, 40, 50, 60, 70\}$. The rows are generated from consecutive random seeds.

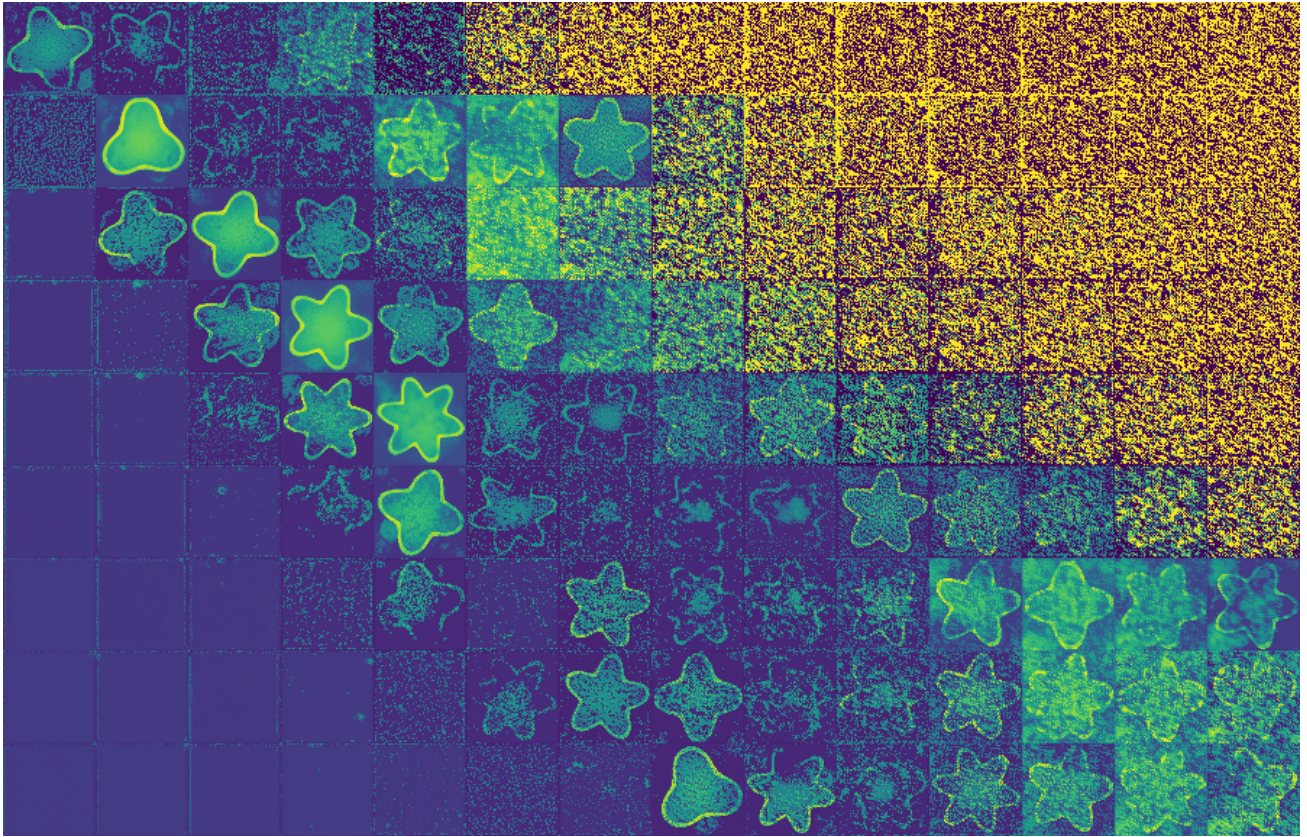


Figure 15. Generated with Algorithm 1 from the same latent noise sample with $\Delta = 0.02$, $p = 8$, and, from left to right, $b \in \{10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75\}$ as well as, from top to bottom, $a \in \{0.0015, 0.002, 0.0025, 0.003, 0.0035, 0.004, 0.0045, 0.005, 0.0055\}$.



Figure 16. Generated with Algorithm 1 from the same latent noise sample with $\Delta = 0.001$, $p = 8$, and, from left to right, $b \in \{0.25, 0.5, 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048\}$ as well as, from top to bottom, $a \in \{0.0005, 0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2\}$.

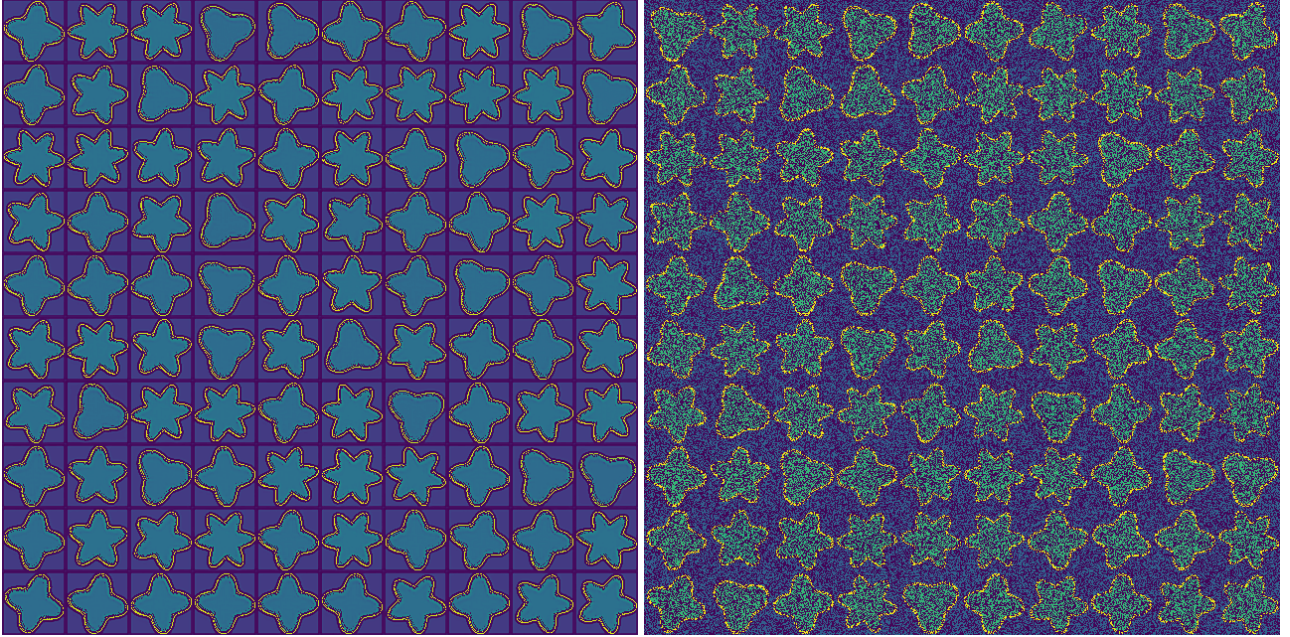


Figure 17. Comparison of generated images using a network trained on our synthetic dataset with pixel removal noise, specifically in a given image every pixel is set to 0 with probability 0.5. Images generated by standard score inference (right). Images generated by extended score inference (left).



Figure 18. Illustrating the inference path for standard score generation (top) compared to extended score generation (bottom), displaying every second image generated during the inference procedure starting at step 20.

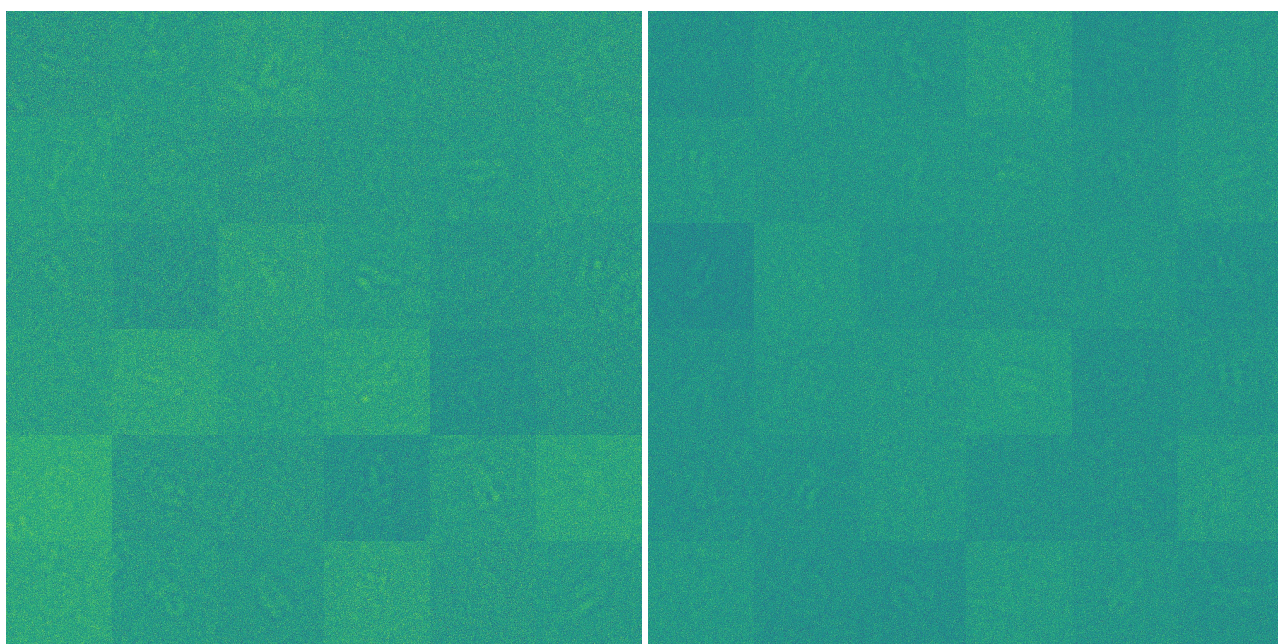


Figure 19. Images from the EMPIAR-11618 dataset (left). Images generated by standard score inference (right).